Grade: Check

Stats, DL27
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Assignment 2

1.

x	fx	xf(x)	$(x-11)^2 f(x)$
20	.05	1	4.05
21	.06	1.26	6
22	.07	1.54	8.47
23	.08	1.84	11.52
24	.09	2.16	15.21
25	.12	3	23.52
26	.15	3.9	33.75
27	.13	3.51	33.28
28	.10	2.8	28.9
29	.10	2.9	32.4
30	.05	1.5	18.15

x Q1: Incorrect standard deviation.

$\boxed{\mu = 25.41 \quad \sigma = 14.66}$ $\sigma^2 = \sqrt{215.15}$

2. a. x Q2a: Incorrect interval probability.

$11 - 14.66 \qquad 11 + 14.66$

$\boxed{47\%}$ $3.66 < \quad < 25.66$ $\quad \overset{20}{.05} + \overset{21}{.06} + \overset{22}{.07} + \overset{23}{.08} + \overset{24}{.09} + \overset{25}{.12} = .47$

✓ Q2b

b. $11 - 2(14.66) \qquad 11 + 2(14.66)$

$\boxed{100\%}$ $-18.32 < \quad < 40.32 = 20 - 30 = 1$

3.

		✓
26	.15	1
27	.13	2
28	.10	3
29	.10	4
30	.05	5

$.15 + .13 + .10 + .10 + .05 = .53 -$

$1 - .53 = \boxed{.47}$ ✓ Q3

✓ Q4

4. $.15 + .13 + .1 + .1 + .05 = .53 = \boxed{53\%}$

5. $1(.15) + 2(.13) + 3(.10) + 4(.10) + 5(.05)$

$.15 + .26 + .3 + .4 + .25 = 1.36$

$1^2(.15) + 2^2(.13) + 3^2(.10) + 4^2(.10) + 5^2(.05)$ ✓ Q5

$.15 + .52 + .9 + 1.6 + 1.25 = 4.42$

$4.42 - (1.36^2) = \sqrt{2.57 = \sigma^2} = \boxed{\sigma = 1.6}$